MAJOR PROJECT SPECIFICATION (IEEE STD 830-1998)

AI-Driven Multimodal Step-Up Authentication System

With Real-Time Deepfake and Voice-Clone Detection for FinTech Security

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Vulnerabilities in Incumbent Auth

Fragility of SMS OTP & Static Biometrics

SMS OTP Attacks: Intercepted easily via SIM-swap fraud, SS7 protocol exploits, and real-time phishing relays without any liveness check.

Static Face ID Spoofing: Vulnerable to simple presentation attacks including printed photos, high-res screen replays, and basic 2D face swaps.

Generative AI Threat Vector

Deepfake Video Synthesis: Generative AI models can now synthesize photorealistic facial likeness in real-time video streams during KYC checks.

Voice Cloning (TTS): Few seconds of audio reference enable attackers to clone victim voices, defeating static voice-biometric systems.

Industry gap: Current systems lack adaptive risk evaluation paired with real-time multimodal liveness verification.

The "Look & Speak" Concept

Adaptive Risk-Driven Challenge

Instead of imposing uniform friction on every transaction, MSAS computes a real-time contextual risk score. Only transactions crossing risk thresholds trigger step-up authentication.

Dual-Modality Active Verification

The user undergoes a simultaneous 8-second visual liveness check ("Look") and dynamic spoken code phrase verification ("Speak") over WebRTC.

Fail-Closed Architecture

Any system timeout, network interruption, or signal ambiguity defaults strictly to Deny or Escalate, ensuring zero silent approvals under failure conditions.



Contextual Risk Evaluation Engine

Transfer Amount

Compares transfer value against account baseline. Abnormally high amounts heavily weigh into cumulative risk.

Geolocation Velocity

Detects impossible speed travel or unverified IP locations relative to historical login hotspots.

Device Fingerprint

Tracks rapid device-switching patterns and browser fingerprint anomalies prior to transfer initiation.

Risk Tiers & Actions

- **Low Risk:** Silent auto-approval without user friction.
- **Medium Risk:** Triggers mandatory "Look & Speak" step-up.
- **High Risk:** Step-up challenge + mandatory admin review.

Time & Payee Context

Inconsistent time-of-day transaction attempts combined with first-time or watchlisted recipient accounts boost score to ensure maximum security for non-routine transfers.

Visual Liveness Core (MediaPipe)

Eye Aspect Ratio (EAR) Blink Detection:

Extracts 468 facial mesh landmarks to monitor involuntary blinking cycles across consecutive video frames.

Randomized Head-Pose Prompts:

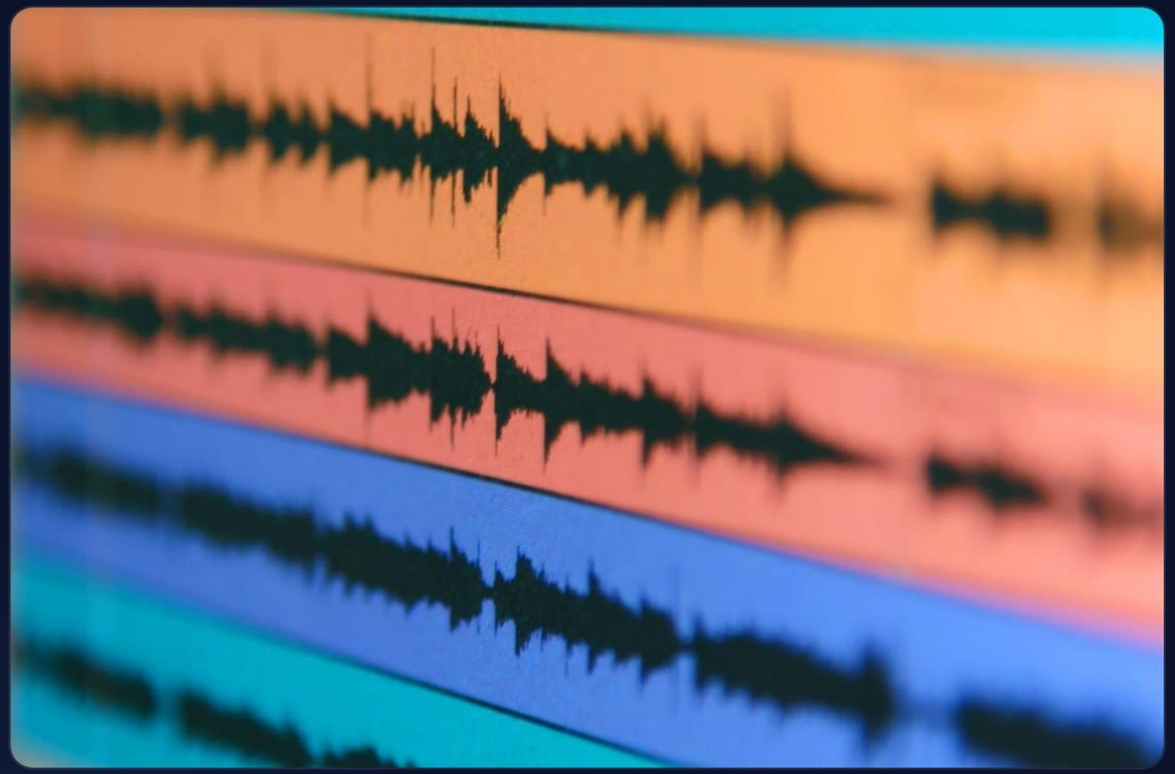
Instructs user to perform dynamic 3D head turns. Defeats flat photos and 2D video replays instantly.

Reticle Alignment Window:

Enforces bounding box alignment, preventing pre-recorded video positioning attacks.



Audio Pipeline & Voice Authenticity



OpenAI Whisper ASR Code Verification

Transcribes spoken audio in real time and matches text against a dynamically generated, single-use phrase cached in Redis. Defeats static voice recording replays.

Voice Authenticity Classifier

Evaluates spectral power distribution, pitch stability, and acoustic phase anomalies to detect synthetic voice clones and TTS engine artifacts.

Multimodal Decision Fusion Rules

VISUAL LIVENESS	SPOKEN CODE	VOICE AUTHENTICITY	RISK TIER	DECISION	SYSTEM RATIONALE
✓ Pass	✓ Match	✓ Authentic	Medium / High	ALLOW	All biometric factors verified; challenge passed cleanly.
✓ Pass	✓ Match	⚠ Flagged	Medium / High	ESCALATE	Possible synthetic voice clone; pending admin review.
✗ Fail	✓ Match	✓ Authentic	Any Tier	DENY	Visual liveness failed; possible photo/video presentation attack.
✗ Fail	✗ Mismatch	✗ Flagged	Any Tier	DENY	Simultaneous failure across modalities; account flagged.

*LLM Risk Reasoning engine generates explainable natural-language logs for all decisions.

System Architecture & Tech Stack

1. Client Layer

Next.js & WebRTC: High-performance frontend capturing zero-latency camera/microphone media streams directly inside the browser.

2. Application Layer

Django REST Framework: Secure API gateway managing authentication sessions, WebRTC tokens, and API orchestration.

3. Intelligence Layer

MediaPipe + Whisper + LLM: On-device vision landmark extraction, Whisper ASR, and decision fusion reasoning engine.

4. Data Layer Architecture

PostgreSQL (persisting user data, transactions & immutable logs) + Redis (caching nonces & session states).

ACID Compliant

Database Schema & Security Posture

Core Relational Tables

TABLE	PRIMARY KEY	KEY RESPONSIBILITIES
`users`	UUID	User profiles, salted password hashes, template pointers.
`transactions`	UUID	Payer/payee info, transfer amounts, risk scores.
`multimodal_audit_logs`	UUID	Immutable decision verdicts, rationale, media hashes.

Privacy & Data Security

Zero Raw Biometric Storage:
Raw photos/audio are destroyed immediately. Only irreversible mathematical embeddings are saved.

Tamper-Evident Audit Logging:
All auth attempts generate durable logs containing cryptographic hashes of verification artifacts.

Data Minimization & Compliance:
Aligned with IT Act 2000 and standard digital banking privacy guidelines.

Quality Attributes & Testing Matrix

Latency Budget

End-to-end "Look & Speak" step-up verification completes under **3.0 seconds**.

Scalability

Supports at least **500 concurrent sessions** with failure rates < 1%.

WCAG 2.1 AA

Captioned prompts, accessible contrast, and non-biometric fallback pathways.

Validation Test Matrix Highlights (IEEE Std 830)

TEST ID	ATTACK SCENARIO / CONDITION	INPUT TEST VECTOR	EXPECTED SYSTEM RESPONSE
TC-03	Photo Spoofing Attack	High-res printed photo held in frame	DENY (No EAR blink / pose movement)
TC-05	Deepfake Voice Clone	Synthetic TTS voice speaking code	ESCALATE (Spectral anomalies flagged)
TC-08	Network Dropout Mid-Check	Dropped WebRTC socket connection	DENY (Fail-Closed policy enforced)

Development Timeline & Phased Roadmap

Month 1 - 3

System Design & SRS: Requirements formulation (IEEE 830), risk scoring logic, and database architecture setup.

Month 4 - 6

Phase 1 Deliverable: Build Risk Engine and Vision Core (MediaPipe liveness). Mid-Term Evaluation & Demo.

Month 7 - 9

Phase 2 Build: Audio pipeline integration (Whisper ASR + Voice Authenticity Classifier) & Decision Fusion Engine.

Month 10 - 12

Testing & Final Viva: Full test matrix execution (TC-01 to TC-10), security hardening, and final presentation.

Thank You!

Questions & Discussion | Snigdha Dwivedi (2023BTCSE018)

Image Sources



https://media.istockphoto.com/id/1166057711/vector/biometric-person-identification-facial-recognition-concept-futuristic-low-polygonal-human.jpg?s=612x612&w=0&k=20&c=bEePF402x94i0sHzgQ9kc8TSGb3jSs4DbeqlusyilPM=

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